SIMATS ENGINEERING
ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1511

COURSE OUTCOME COVERED
CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**
Question Count: 5

Code of Conduct
I _S.Amulya(192321044)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: ____________________

Assessment Tasks
Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

1. Media Company Migrating Video Streaming Platform to the Cloud (10 Marks)

Introduction

A media company requires a cloud architecture capable of streaming videos to millions of concurrent users with minimal buffering. The solution must be highly scalable, fault tolerant, and cost-effective.

Cloud Architecture Design

The proposed architecture includes:

- **Users** access the platform through the Internet.
- **Content Delivery Network (CDN)** caches videos at edge locations close to users.
- **Load Balancer** distributes incoming traffic across multiple application servers.
- **Auto-Scaling Group** automatically adds or removes servers based on traffic.
- **Object Storage** stores video files.
- **Cloud Database** stores user profiles, subscriptions, and viewing history.
- **Monitoring Service** tracks CPU, memory, bandwidth, and latency.

Scalability and Caching

- Auto-scaling launches additional servers during peak demand.
- CDN caches frequently viewed videos at edge servers.
- In-memory caching stores popular metadata and user sessions.
- These techniques reduce server load and buffering.

Load Balancing and Auto-Scaling

- Load balancer evenly distributes requests among healthy servers.
- Health checks remove failed servers automatically.
- Auto-scaling responds to CPU usage, network traffic, or request count.
- Users experience uninterrupted video playback even during peak traffic.

Cost Optimization Strategies

- Use auto-scaling to avoid paying for idle resources.
- Store old videos in low-cost archive storage.
- Use spot/preemptible instances for background video processing.
- Compress video files to reduce storage and bandwidth costs.
- CDN reduces bandwidth charges from the origin server.

Conclusion

A combination of CDN, load balancing, auto-scaling, and cloud storage enables high-performance video streaming while minimizing buffering and optimizing operational costs.

2. Fintech Startup Using Containers Across Multiple Clouds (10 Marks)

Introduction

A fintech startup deploys applications across multiple cloud providers to improve availability and avoid vendor lock-in. Managing containers consistently across clouds requires orchestration.

Proposed Solution

- Package applications using **Docker containers**.
- Deploy containers using **Kubernetes** across multiple cloud providers.
- Use a centralized container registry.
- Implement CI/CD pipelines for automated deployment.

Container Orchestration

Kubernetes provides:

- Automated deployment
- Auto-scaling
- Load balancing
- Self-healing
- Rolling updates
- Resource scheduling

This ensures consistent deployments across cloud providers.

Service Discovery

Kubernetes automatically assigns DNS names to services.

- Containers communicate using service names instead of IP addresses.
- Failed containers are replaced automatically.
- Requests are routed to healthy service instances.

Scaling

Horizontal Pod Autoscaler increases or decreases container replicas based on CPU or memory utilization. Cluster Autoscaler adds or removes worker nodes when required.

Benefits of Multi-Cloud

- High availability
- Disaster recovery
- Vendor independence
- Better geographic coverage
- Regulatory compliance

Risks

- Increased management complexity
- Higher networking costs
- Security policy differences
- Data synchronization challenges
- More complex monitoring

Conclusion

Using Kubernetes with multi-cloud deployment provides scalability, portability, and resilience while reducing vendor lock-in, although careful management is required.

3. Backup and Disaster Recovery for an E-Learning Platform (10 Marks)

Introduction

An e-learning platform suffers data loss due to accidental deletion. A strong backup and disaster recovery strategy is required to restore services quickly.

Proposed Solution

Cloud-Native Backup

- Schedule automatic daily backups.
- Store backups in cloud object storage.
- Encrypt backup files.

Replication

- Replicate data across multiple availability zones or regions.
- Maintain synchronized backup copies.

Versioning

- Enable object versioning.
- Deleted or modified files can be restored from previous versions.

Recovery Time Objective (RTO)

RTO defines the maximum acceptable downtime.

To achieve low RTO:

- Maintain standby systems.
- Automate restoration.
- Use replicated infrastructure.

Recovery Point Objective (RPO)

RPO defines the maximum acceptable data loss.

To achieve low RPO:

- Perform frequent backups.
- Enable continuous replication.
- Use database transaction logs.

Business Continuity Steps

1. Detect system failure.
2. Switch to replicated resources.
3. Restore latest backup.
4. Verify data integrity.
5. Resume application services.
6. Continue monitoring until normal operation.

Conclusion

Cloud-native backup, replication, and versioning provide reliable disaster recovery while minimizing downtime and data loss.

4. Hybrid Cloud Architecture for a Government Agency (10 Marks)

Introduction

A government agency stores highly sensitive citizen data that must remain on-premises while using public cloud services for non-sensitive applications.

Hybrid Cloud Architecture

- Sensitive applications remain in the on-premises data center.
- Public cloud hosts websites, portals, and analytics.
- Secure VPN or dedicated private connection links both environments.
- Centralized monitoring manages both infrastructures.

Secure Communication

- VPN or dedicated cloud connection encrypts network traffic.
- Firewalls filter unauthorized access.
- Network segmentation separates sensitive and non-sensitive workloads.

Identity Management

- Single Sign-On (SSO)
- Multi-Factor Authentication (MFA)
- Identity Federation
- Role-Based Access Control (RBAC)

These provide consistent authentication across both environments.

Encryption

- Encrypt stored data using AES-256.
- Encrypt network communication using TLS/SSL.
- Secure encryption key management.

Challenges

- Managing multiple environments
- Higher operational complexity
- Data synchronization
- Compliance monitoring
- Increased integration effort

Conclusion

Hybrid cloud allows government agencies to maintain regulatory compliance while benefiting from cloud scalability through secure communication, strong identity management, and encryption.

5. Multi-Region High Availability for a SaaS Provider (10 Marks)

Introduction

A SaaS provider requires continuous service availability even if an entire cloud region becomes unavailable.

Multi-Region Deployment Strategy

- Deploy application servers in multiple geographic regions.
- Use global load balancing.
- Maintain identical application environments.
- Distribute traffic to the nearest healthy region.

Data Replication

- Use synchronous replication within the same region for consistency.
- Use asynchronous replication across regions for disaster recovery.
- Maintain regular database backups.

Failover Mechanism

- Health checks continuously monitor regional services.
- If one region fails, traffic automatically switches to another healthy region.
- DNS or global load balancers perform automatic failover.

Monitoring and Alerting

Cloud monitoring tools continuously monitor:

- CPU utilization
- Memory usage
- Network traffic
- Application latency
- Database performance
- Regional health

Alerts are sent immediately through email, SMS, or messaging services when abnormalities are detected.

Benefits

- High availability
- Fault tolerance
- Business continuity
- Reduced downtime
- Improved customer satisfaction

Conclusion

A multi-region architecture with data replication, automatic failover, and continuous monitoring ensures uninterrupted SaaS services even during regional outages, providing high availability and reliable disaster recovery.